Coalition Lattices for Interpretable Machine Learning

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In interpretable machine learning, breakdown plots and Shap-values can be seen as different methods applied to a so-called coalition lattice. In these notes, I show how to create a coalition lattice and how we can derive different post-hoc interpretations from that lattice. Specifically, can ask what each predictive feature's importance is for the overall prediction. Throughout, I work with a simple example of an ensemble learner.

Importantly, our focus is on **local interpretation**. We evaluate the coalition lattice for an instance with specific feature values. This sets our discussion apart from methods of global interpretation such as feature importance metrics or partial dependence plots.

1 Example

1.1 An Ensemble of Five Trees

Imagine we have three binary features, $A, B, C \in \{0, 1\}$. We grow 5 trees with those features, which are shown in Figure 1. Each tree produces a set of predicted values, $\hat{y}$, at the leafs. The fraction of training instances that belong to a certain leaf is called the sample weight and is denoted by w.

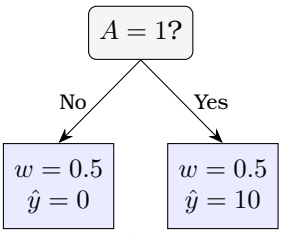
The ensemble learner simply averages across the predictions. Given a vector $\boldsymbol{x}$ of feature values—in our example, 0s and 1s—let $T_m(\boldsymbol{x})$ denote the prediction in the mth tree. Then the ensemble prediction across the 5 trees is

$$f(\boldsymbol{x}) = \frac{1}{5} \sum_{m=1}^{5} T_m(\boldsymbol{x}) \tag{1}$$

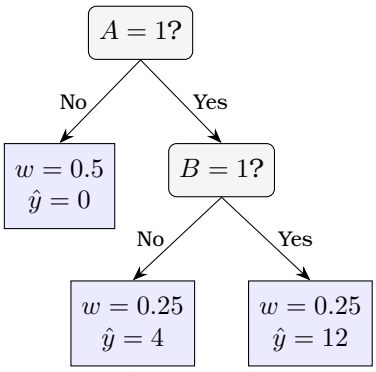
1.2 Predictions from the Trees and Ensemble

For each tree, we can generate a series of predictions. For our set of trees, those can be broken down into four types:

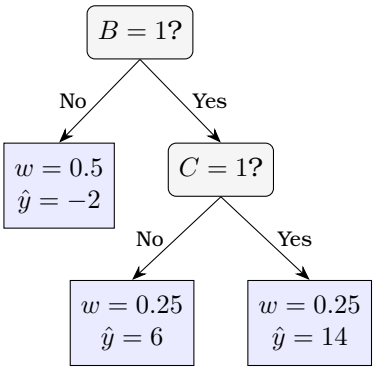
Figure 1: Five Hypothetical Candidate Trees



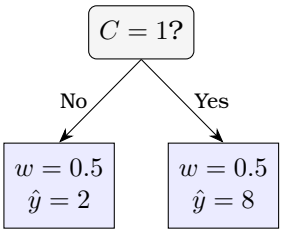
(a) Split on A



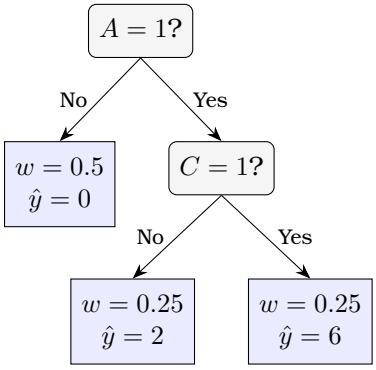
(b) Split on A, then B



(c) Split on B, then C



(d) Split on C



(e) Split on A, then C

Note: Each tree makes splits on up to two features from the set $\{A, B, C\}$. The parameter w indicates the fraction of cases residing in a particular leaf. The outcome $\hat{y}$ is the predicted score at a particular leaf.

1. $v(\emptyset)$ is the baseline prediction, i.e., the prediction we make when ignoring feature information. This is simply the weighted average of the predictions at each of the leafs.

2. Conditioning on a single feature. The prediction is based on $\{A\}$, $\{B\}$, or $\{C\}$ by itself. We assume knowledge only on this feature. The predictions are denoted as $v(\{A\})$, $v(\{B\})$, and $v(\{C\})$, respectively.

3. Conditioning on a pair of features. We assume knowledge about the values of a feature pair. This generates predictions $v(\{A, B\})$, $v(\{A, C\})$, and $v(\{B, C\})$.

4. $v(\{A, B, C\})$ is the prediction based on the full set of feature values. We assume we know the values on all of the features.

To see how the predictive process unfolds, let us consider an instance that scores 1 on all three features. We denote the feature vector of this instance by $\boldsymbol{x}^*$. Let us now proceed through the second tree where we first split on A and then on B.

Baseline Prediction If we ignore all of the information in $\boldsymbol{x}^*$, then our prediction is the average across the $\hat{y}$s in the three terminal leafs of the tree. This is a weighted average, where we take the w at each leaf into account. Hence,

$$v(\emptyset) = 0.5 \cdot 0 + 0.25 \cdot 4 + 0.25 \cdot 12 = 4.0$$

Conditioning on a Single Feature If our interest is in $v(\{A\})$, then we consider the first element of $\boldsymbol{x}^*$. Since $A = 1$, we end up in the right branch of the tree. We condition solely on A, so that we are agnostic about the consequences of B, the next feature that generates a split. We average across the predictions from the second split, which are 4 and 12, respectively. Those possibilities occur with equal probability, so that

$$v(\{A\}) = 0.5 \cdot 4 + 0.5 \cdot 12 = 8.0$$

We can also consider $v(\{B\}$. If we assume that only B is known, then we do not know whether $B = 1$ co-occurs with $A = 0$ or $A = 1$. Thus, we give half of the probability to $\hat{y} = 0$ ($A = 0$). Should A happen to be 1, then we automatically know that we need to follow the right branch, since $B = 1$. Thus, the prediction becomes

$$v(\{B\}) = 0.5 \cdot 0 + 0.5 \cdot 12 = 6.0$$

What about $v(\{C\})$? Since C is not in the second tree, conditioning on it does not add any information relative to $v(\emptyset)$. Hence,

$$v(\{C\}) = 4.0$$

Table 1: Predictions from the Five Regression Trees

		Known features							
Tree	Structure	$\emptyset$	$\{A\}$	$\{B\}$	$\{C\}$	$\{A, B\}$	$\{A, C\}$	$\{B, C\}$	All
T_1	Split on A (one level)	5.0	10.0	5.0	5.0	10.0	10.0	5.0	10.0
T_2	Split on A, then B if $A = 1$	4.0	8.0	6.0	4.0	12.0	8.0	6.0	12.0
T_3	Split on B, then C if $B = 1$	4.0	4.0	10.0	6.0	10.0	6.0	14.0	14.0
T_4	Split on C (one level)	5.0	5.0	5.0	8.0	5.0	8.0	8.0	8.0
T_5	Split on A, then C if $A = 1$	2.0	4.0	2.0	3.0	4.0	6.0	3.0	6.0
Ensemble	Average over the five trees	4.0	6.2	5.6	5.2	8.2	7.6	7.2	10.0

Notes: The table shows all eight possible configurations of conditionals, the predictions in each tree, as well as the ensemble predictions. The predictions are for an instance with $A = B = C = 1$.

Conditioning on a Pair of Features There are three possible pairs of features.

1. $\{A, B\}$: We arrive at a deterministic prediction. We know that $A = 1$ and $B = 1$, so that follow the right branch on A and then the right branch on B. The two right turns yield a deterministic prediction instead of an average:

$$v(\{A, B, C\}) = 12.0$$

2. $\{A, C\}$: While we split on A, we do not on C. The score on A puts us in the right branch. We do not know B, so we average over the last two leafs, giving each equal weight:

$$v(\{A, C\}) = 0.5 \cdot 4 + 0.5 \cdot 12 = 8.0$$

3. $v(\{B, C\})$: We know that $B = 1$, but we do not know the value of A. There is a 0.5 weight on $\hat{y} = 0$ ($A = 0$) and a 0.5 weight on $\hat{y} = 12$ ($A = 1$ in conjunction with the knowledge that $B = 1$). Hence,

$$v(\{B, C\}) = 0.5 \cdot 0 + 0.5 \cdot 12 = 6.0$$

Conditioning on All of the Features Now consider the prediction on the full set of features. We know that $A = 1$ and $B = 1$, so that follow the right branch on A and then the right branch on B. The two right turns yield a deterministic prediction instead of an average:

$$v(\{A, B, C\}) = 12.0$$

Table 1 repeats the process from the second tree for all trees that were fitted. I leave the verification of the numbers as an exercise. The ensemble simply averages across the predictions in each column.

2 The Coalition Lattice

The coalition lattice represents every possible combination of features and shows how a model's output changes as features come together (whence the term "coalition"). We also call this the game-theoretic hypercube. In our running example, this hypercube operates on the Boolean subset lattice $\{0,1\}^3$.

2.1 Generating the Lattice

Figure 2 shows the game-theoretic hypercube for the ensemble of 5 trees over 3 features. The front-left bottom corner shows the baseline prediction of the ensemble (4.0). Moving to the front-right bottom, we obtain the ensemble prediction due to feature A if it is added by itself. Moving to the front-left top corner, we see the prediction due to B by itself, whereas the front-right top corner shows the joint prediction due to A and B. The back-panel shows the predictions due to C, the combination of A and C, the combination of B and C, and the combination of A, B, and C.

The specific predictions in the lattice reflect the specific instance we chose, namely one where $A = B = C = 1$. The architecture of the cube itself, however, is invariant across instances. It simply captures the full range of constellations in which features exert their effects.

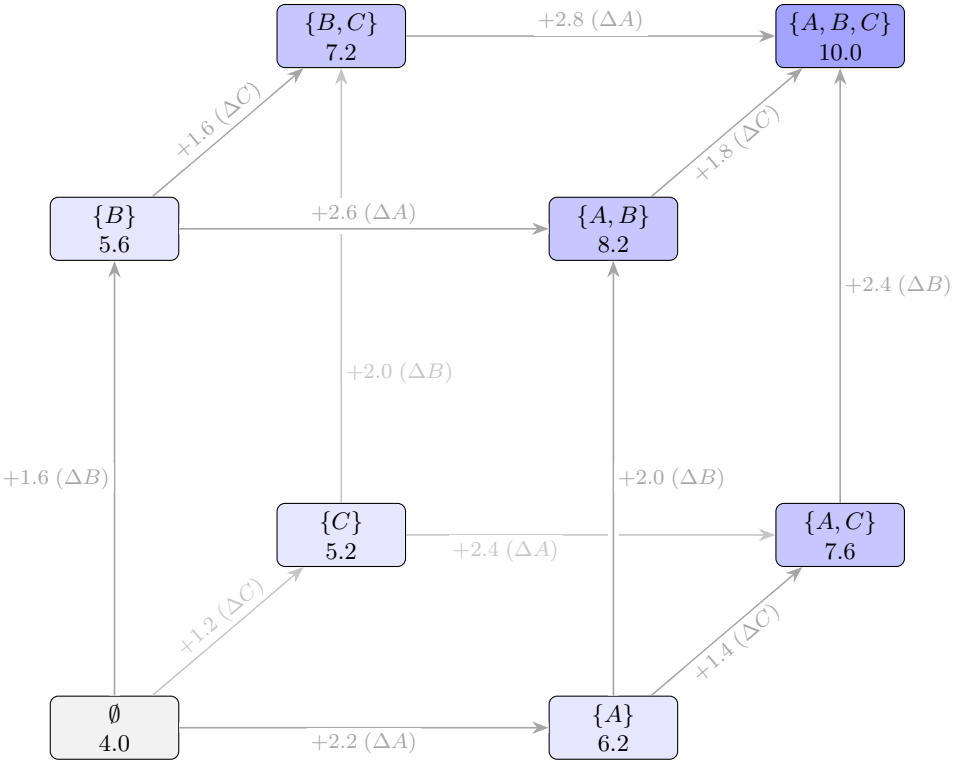
2.2 Interpreting the Edges

The lattice edges provide important information in the form of the Δs. For instance, moving from node $\emptyset$ to node $\{A\}$, increases the prediction from 4.0 to 6.2. This results in $\Delta A = 2.2$. We can think of this as the **marginal effect** of A when it is added on its own.

A key aspect of the Δs is that they allow us to assess the **combinatorial context** of an effect. Effects are rarely fixed but depend on which other features are already present. Take for instance the effect of C. Moving from $\emptyset$ to $\{C\}$ yields $\Delta C = 1.2$. If instead we move from $\{A\}$ to $\{A, C\}$, we observe $\Delta C = 1.4$. Thus, the effect of C is bigger when A is already present than when we start with an empty set and add C as the first feature to generate a split. At least, that is true for the instance where $A = B = C = 1$. Notice that the two edges for Δ_C is that they run parallel.

The Δs allow us to **quantify interactions**. If features are purely additive, the marginal jumps $\Delta_{\cdot}$ are constant across the lattice. Any variation around parallel edges indicates an interaction or synergy. For instance, we can assess the impact

Figure 2: The Subset Lattice as a Cube



Notes: Each vertex is a conditioning set with its expected prediction; each edge is the marginal contribution of adding one feature. Shading darkens with the number of known features, from the baseline $\emptyset$ to the full set $\{A, B, C\}$.

of the presence of B on the marginal effect of A by comparing the edge from $\emptyset \to \{A\}$ to that from $\{B\} \to \{A, B\}$. We call the first edge $\Delta_A(\emptyset)$ and the second one $\Delta_A(\{B\})$. The first edge shows the marginal effect of introducing A as the first feature, whereas the second edge shoes the effect of introducing A after B. The interaction can now be decomposed as

$$\begin{aligned} \text{Interaction}(A, B) &= \Delta_A(\{B\}) - \Delta_A(\emptyset) \\ &= v(\{A, B\}) - v(\{A\}) - v(\{B\}) + v(\emptyset) \end{aligned}$$

For the instance with $A = B = C = 1$, this results in an interaction effect of 0.4.

2.3 One Plot, Different Questions

The plot and its edges allow us to answer very different questions about the predictions.

1. Breakdown plots ask: What is the impact of features along one specific path of addition?
2. SHAP asks: What is the average contribution of a feature across all possible combinatorial contexts?

3 Breakdown Plots

Breakdown plots select exactly one path in the lattice, moving from the empty set to a full consideration of all features. They do so using a simple heuristic: look for the largest Δ. In this sense, the breakdown plot chooses greedy paths through the lattice.

3.1 Additive Breakdown Plots

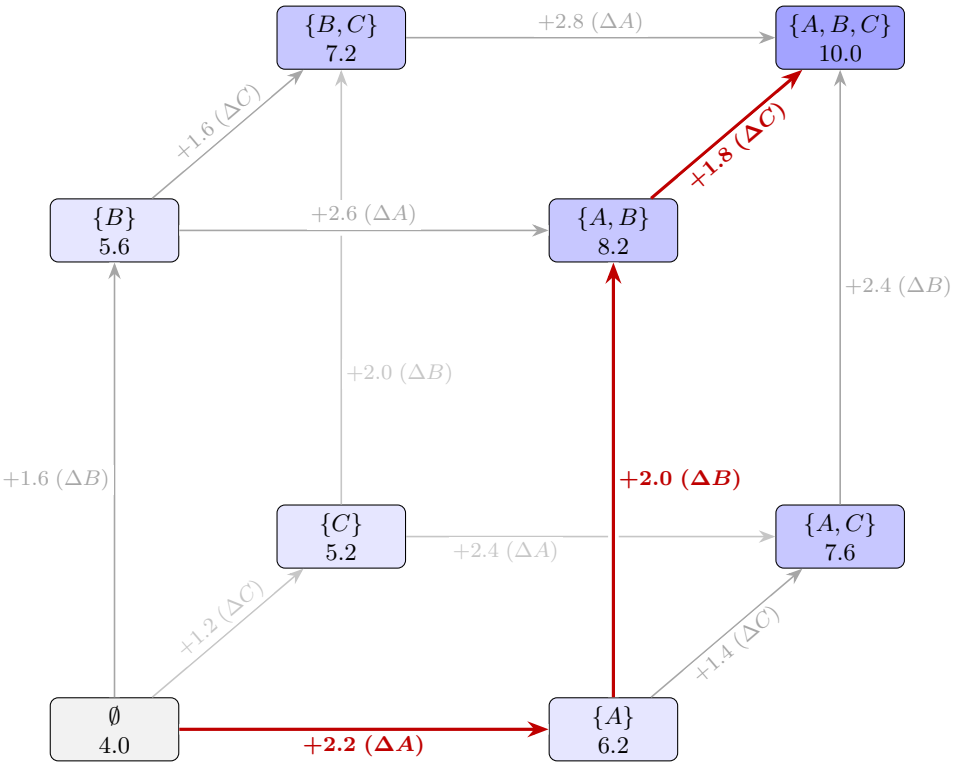
In an additive breakdown plot, we can move only along the edges. Within this constraint we select the steepest Δs to determine the sequence of movements (Biecek & Burzykowski, 2021; Staniak & Biecek, 2019).

How does this work for Figure 2? Starting at the empty set-node, we can add A, B, or C as the first feature. From Figure 2, we know that $\Delta_A(\emptyset) = 2.2$, $\Delta_B(\emptyset) = 1.6$, and $\Delta_C(\emptyset) = 1.2$. The biggest jump comes from adding A.

In the next step, the question is whether to add B or C. We see that $\Delta_B(\{A\} = 2.0$ and that $\Delta_C(\{A\}) = 1.4$. The greedy heuristic indicates that we should add feature B next.

With only three binary features, the next step is trivial: we add feature C. $\Delta_C(\{A, B\}) = 1.8)$. Figure 3 shows the greedy path through the lattice in red. The path takes us

Figure 3: Greedy Path Through a Coalition Lattice

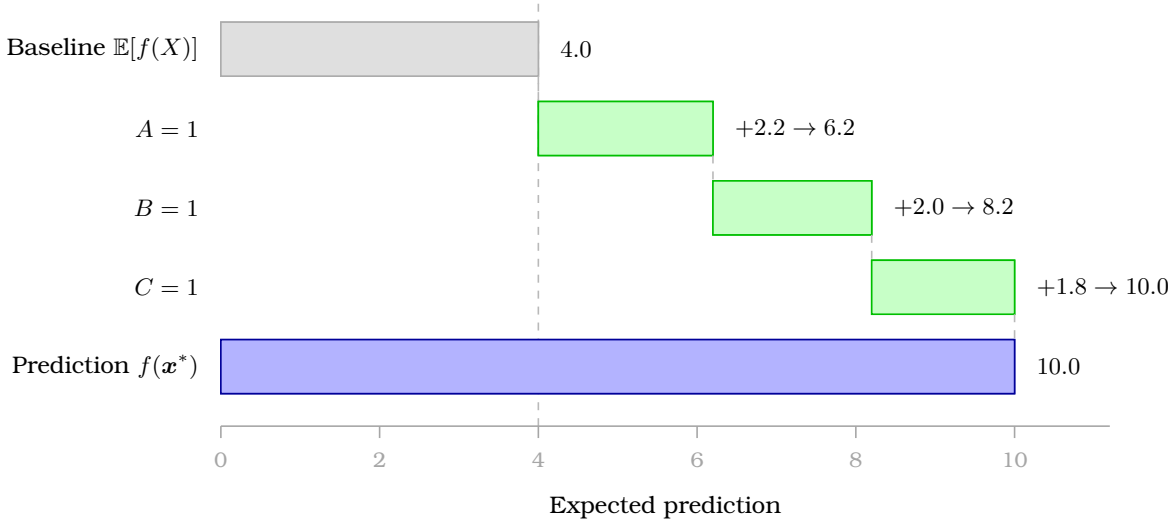


Notes: Features are added in the order A, then B, then C based on the biggest increase in the predicted value (greedy path).

to a final prediction of 10.

The breakdown plot shows the accumulation of effects to the final prediction (Figure 4). We start with the baseline ($\emptyset$) of 4.0. We then add the marginal effect of A to boost the predicted value to 6.2. Adding B in the next step further boosts the prediction to 8.2. Finally, adding C in the third step, adds the final 1.8 points that gets us to the prediction value of 10.0.

Figure 4: Additive Breakdown Plot



Notes: Each bar is the marginal contribution of adding one feature to the conditioning set, following the highlighted path through the coalition lattice. The contributions telescope from the baseline 4.0 to the prediction 10.0.

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3.2 Interactive Breakdown Plots

Gosiewska and Biecek (2019) caution against the use of additive explanations, which includes additive breakdown plots, when there are interactions. Indeed, there are two problems with the additive breakdown plot when an effect is truly the joint influence of a set of features.

1. The explanation path may be unnecessarily complex. For instance, if the action is truly in the joint presence of A and B, then why visit B only after visiting A? Why not immediately draw a path to $\{A, B\}$? That would require us drawing a vector from the $\emptyset$-node to $\{A, B\}$, which is not allowed in an additive breakdown plot. But that is just a constraint from this specific method and not something inherent in the data generating process.

2. A feature that came second may appear more important because it ends up carrying the interaction penalty or bonus. This can induce a bias in feature importance.

iBreakDown makes it possible to directly move to the interaction instead of passing through the constituent components (Gosiewska & Biecek, 2019). This means that we give up on the restriction that we must move along the edges of the lattice, i.e., use vectors at angles of 0 and 90 degrees. Instead, we can draw diagonal vectors. The single constraint is that vectors can only move to supersets. Thus moving from $\{A\}$ to $\{A, B, C\}$ is allowed, but moving from $\{A\}$ to $\{B, C\}$ is not. In practice, this means that the vectors must be located on the sides of the hypercube and cannot be located in the interior. We call this face diagonal traversal.

In its crudest form—we will add some nuance to this momentarily—greediness means that we go directly to the place where the Δ relative to the empty set is greatest. We have already seen that $\Delta_A(\emptyset) = 2.2$, $\Delta_B(\emptyset) = 1.6$, and $\Delta_C(\emptyset) = 1.2$. But we are no longer limited to these choices. We can also go to $\{A, B\}$, $\{A, C\}$, or $\{B, C\}$, keeping the face diagonal traversal constraint in mind. Figure 2 shows $\Delta_{A,B}(\emptyset) = 8.2$, $\Delta_{A,C}(\emptyset) = 7.6$, and $\Delta_{B,C}(\emptyset) = 7.2$. Of all the possibilities for leaving the empty set, including adding a single feature, introducing the interaction between A and B brings us the biggest change. Hence, we would make the following move: $\emptyset \to \{A, B\}$. The next move would then be $\{A, B\} \to \{A, B, C\}$. The moves are illustrated in Figure 5.

Comparing Figures 3 and 5, we clearly see that we saved one move. Instead of $\emptyset \to \{A\} \to \{A, B\} \to \{A, B, C\}$, we took the path $\emptyset \to \{A, B\} \to \{A, B, C\}$. This means that the breakdown plot consists of four instead of five bars, as is shown in Figure 6.

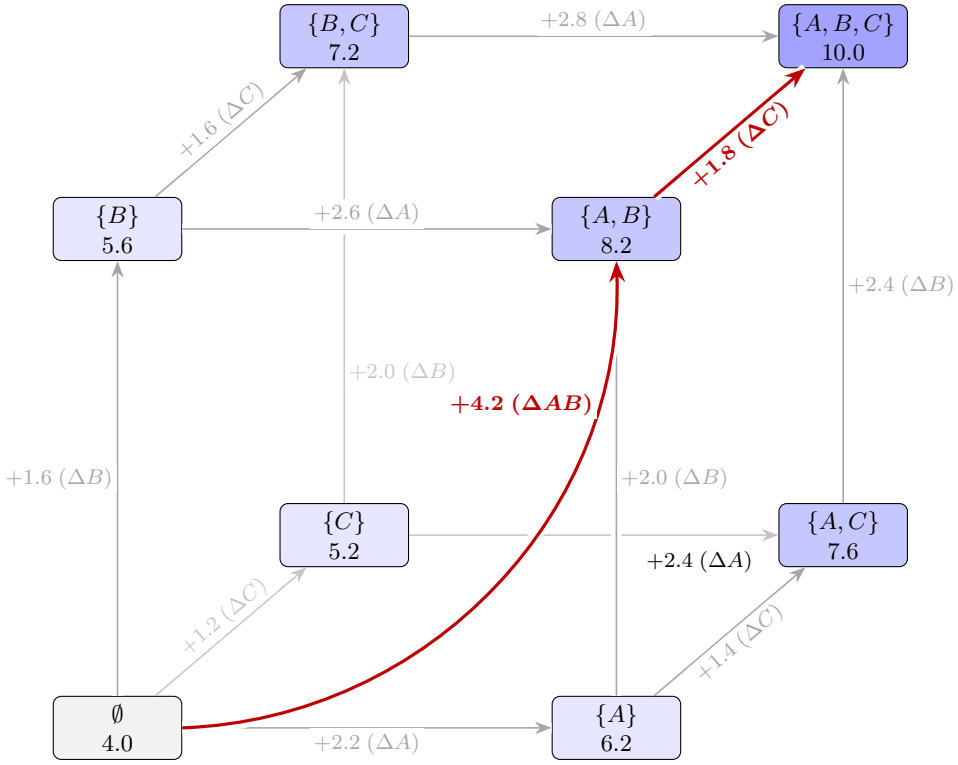
In constructing Figure 6, we kept A and B together because there joint effect exceeds their additive effect by 0.4. However, Gosiewska and Biecek (2019) give considerable leeway in controlling the shape of the breakdown plot through a parameter they call **interaction preference** (also see Biecek & Burzykowski, 2021). Depending on how this is set, we could still end up with Figure 4.

The key to understanding interaction preference is the deviation from additivity. Given two features i and j,

$$\Delta^I_{\{i,j\}}(\boldsymbol{x}^*) = \Delta_{\{i,j\}|\emptyset}(\boldsymbol{x}^*) - \underbrace{\left[\Delta_{\{i\}|\emptyset}\boldsymbol{x}^*) + \Delta_{\{j\}|\emptyset}(\boldsymbol{x}^*)\right]}_{\text{Additivity}} \tag{2}$$

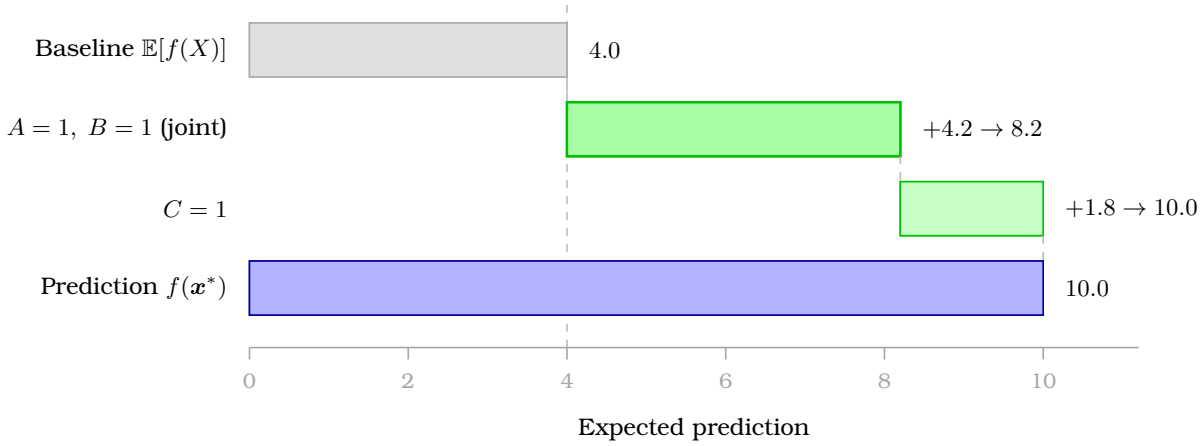
In our example, the additive contribution of A and B is $2.2 + 1.6 = 3.803.8$. However, the interaction effect adds 4.2 relative to the baseline. The difference of 0.4 is the deviation from additivity.

Figure 5: iBreakDown Paths



Notes: Allowing for face diagonal traversal, we move from $\emptyset$ to $\{A, B\}$ directly. We then move to adding C in a second step.

Figure 6: Interactive Breakdown Plot



Notes: iBreakDown plot keeping A and B together. The joint contribution of 4.2 exceeds the sum of the separate contributions, $2.2 + 1.6 = 3.8$, by 0.4; that interaction is why the pair is not split.

So far, this all points to going the way of $\emptyset \rightarrow \{A, B\}$, as we showed in Figure 5. However, interaction preference scales the ranking. The trick is that it does so only for the interaction. Single features are scored by their absolute contribution:

$$\text{Score}(i) = |\Delta_{\{i\}|\emptyset}(\boldsymbol{x}^*)| \tag{3}$$

However, pairs of features are scored by their interaction deviation multiplied by the interaction preference, γ:

$$\text{Score}(i, j) = \gamma \cdot |\Delta^I_{\{i,j\}}(\boldsymbol{x}^*)| \tag{4}$$

Here, $\gamma \geq 0$. The default in iBreakDown is $\gamma = 1$. If $\gamma > 1$, then the interaction gets a boost, making it more likely that features will be considered as pairs rather than individually. If $\gamma = 0$, then interactive breakdown reverts to additive breakdown.

In the formulation with interaction preference, there is a bias toward selecting individual features unless γ is large. The reason is that the scores are sorted by magnitude and the highest score is picked using greediness. In our example, $\text{Score}(A) > \text{Score}(B) > \text{Score}(C) > \text{Score}(A, B)$, unless $\gamma > 5.5$. That means that small values of γ bias the interactive breakdown plot in the direction of additive breakdown. Only if the interaction deviation is very large or if we care deeply about joining features, will we choose a face diagonal traversal.

3.3 Adding Uncertainty

In their pure form, breakdown plots do not convey any uncertainty about feature importance for a local prediction. There is one greedy path through the coalition hypercube and this informs the plot.Gosiewska and Biecek (2019) added uncertainty by generating m random samples of variable orders, thus destroying greediness.[1] If we stay within the framework of additive breakdown plots, the benefit of this approach over Shapley-values is unclear. There would be a pay-off for interactive breakdown plots because they follow a different logic from Shapley values.

3.4 Evaluation

The greedy approach has the advantage of simplifying the combinatorics of a coalition hypercube. Of the many paths that are possible in the lattice, it picks just one to navigate. However, in the process of doing so, the entire coalition logic of the lattice is destroyed.

Is there a justification for this flagrant neglect of coalitions? The approach is in line with the take-the-best heuristic developed by Gigerenzer and Goldstein (1996) and elimination-by-aspects Tversky (1972). It may also reflect the actual practice of evidence triage (e.g., Green & Mehr, 1997). Practitioners often ask these kinds of questions:

[1] Herein lies the confusion from the lecture on interpretable machine learning. There I described the sampling, but did not show it.

- What is the biggest red flag?
- Given that red flag, what compounds it most?
- What residual factors remain?

If these are the questions one would like to answer, then breakdown plots are a reasonable choice.

4 SHAP

SHAP-values are named after Shapley (1953), who studied cooperative game theory. In this branch of formal modeling, a set of actors comes together to produce a good through cooperation. Shapley asked what a fair recognition of each actor's contribution looks like, keeping in mind that the order in which the actors make their contribution matters. Imagine for instance 3 actors who build a house. A is the architect, B is the contractor, and C is the interior designer. It stands to reason that A's contribution is at its most useful if it comes first. But it does not necessarily have to be that way. Perhaps C's ideas inspire the architect to draw up different plans, so that A may do better work when they have seen C's ideas. Shapley argued that we must consider all possible orders of the three actors, measure each actor's contribution in each order, and then average across orders to get a fair sense of each actor's contribution.

What does this have to do with machine learning? Lundberg and Lee (2017) took the idea to assess feature importance. If instead of treating A, B, and C as actors, we view them as different predictive features, then Shapley's approach can be used to assess each feature's contribution across different orderings of those features.

4.1 TreeSHAP

Starting with the empty set, classical SHAP traverses every possible path in the coalition lattice that still satisfies the superset principle (see interactive breakdown plots). With 3 features there are $3! = 6$ such paths. More generally, it is $P!$. For each path, we compute the marginal step (i.e., contribution) of each feature. At the end, we use the average of these marginal steps as a measure of feature importance. This is called the SHAP-value for that feature. Naturally, we also obtain information about the spread in the contribution per feature. This provides a sense of uncertainty about each feature's importance.

For our example, the 6 paths are as follows: (1) $\emptyset \to \{A\} \to \{A, B\} \to \{A, B, C\}$; (2) $\emptyset \to \{A\} \to \{A, C\} \to \{A, B, C\}$; (3) $\emptyset \to \{B\} \to \{A, B\} \to \{A, B, C\}$; (4) $\emptyset \to \{B\} \to \{B, C\} \to \{A, B, C\}$; (5) $\emptyset \to \{C\} \to \{A, C\} \to \{A, B, C\}$; and (6) $\emptyset \to \{C\} \to \{B, C\} \to \{A, B, C\}$. Table 2 shows the partial effects of each feature along a path—the marginal steps. The final row averages the values across the 6 paths. These

Table 2: A TreeSHAP Example

		Marginal Steps		
Index	Permutation	A	B	C
1	$\emptyset \to \{A\} \to \{A,B\} \to \{A,B,C\}$	$\Delta_A(\emptyset) = 2.2$	$\Delta_B(\{A\}) = 2.0$	$\Delta_C(\{A,B\}) = 1.8$
2	$\emptyset \to \{A\} \to \{A,C\} \to \{A,B,C\}$	$\Delta_A(\emptyset) = 2.2$	$\Delta_B(\{A,C\}) = 2.4$	$\Delta_C(\{A\}) = 1.4$
3	$\emptyset \to \{B\} \to \{A,B\} \to \{A,B,C\}$	$\Delta_A(\{B\}) = 2.6$	$\Delta_B(\emptyset) = 1.6$	$\Delta_C(\{A,B\}) = 1.8$
4	$\emptyset \to \{B\} \to \{B,C\} \to \{A,B,C\}$	$\Delta_A(\{B,C\} = 2.8$	$\Delta_B(\emptyset) = 1.6$	$\Delta_C(\{B\}) = 1.6$
5	$\emptyset \to \{C\} \to \{A,C\} \to \{A,B,C\}$	$\Delta_A(\{C\}) = 2.4$	$\Delta_B(\{A,C\}) = 2.4$	$\Delta_C(\emptyset) = 1.2$
6	$\emptyset \to \{C\} \to \{B,C\} \to \{A,B,C\}$	$\Delta_A(\{B,C\}) = 2.8$	$\Delta_B(\{C\}) = 2.0$	$\Delta_C(\emptyset) = 1.2$
Average		$\phi_A = 2.5$	$\phi_B = 2.0$	$\phi_C = 1.5$

Notes: The table shows all six paths from $\emptyset$ to $\{A, B, C\}$ and the partial effects of each feature along that path. Those effects are for an instance with $A = B = C = 1$.

are the **SHAP values**. Notice that adding the SHAP values to the baseline predictions yields the predicted value of 10.

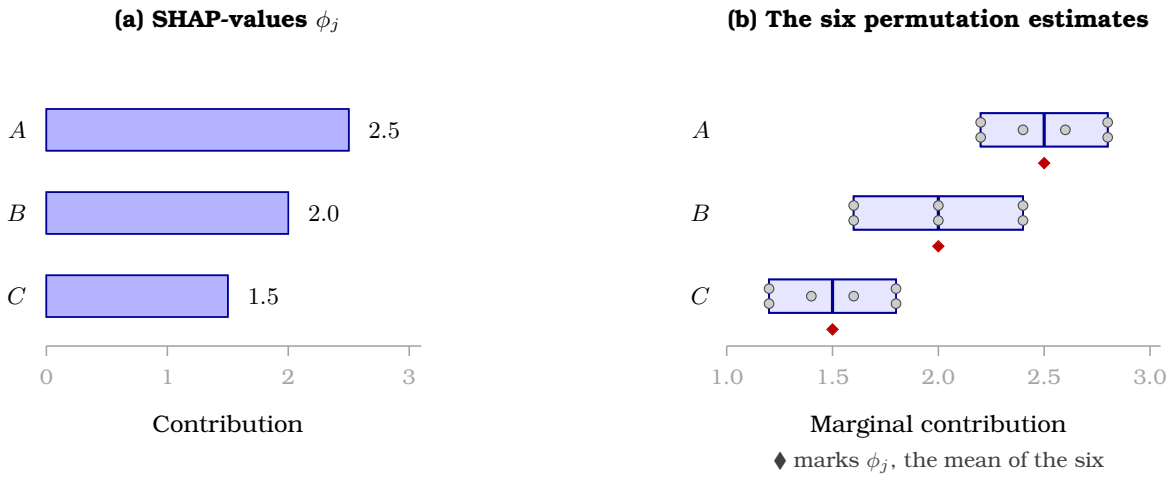
Figure 7 shows the SHAP-values for our example, as well as the permutation estimates. The latter are shown in the form of a box plot that shows the spread as the inter-quartile range. We observe that the marginal steps of B varies the most, ranging from 1.6 to 2.4. Those for A range between 2.2 and 2.8, while those for C have a similar range (1.2 to 1.8).

4.2 Asymmetric SHAP-Values

In causal inference, it may not make sense to be using TreeSHAP because we have a theory of the causal process that is commensurate with only some of the paths in the lattice. For instance, imagine that our theory is captured by the directed acyclical graph (DAG) in Figure 8. Here, A is the antecedent of B; B is a mediator. C is an exogenous, independent covariate.

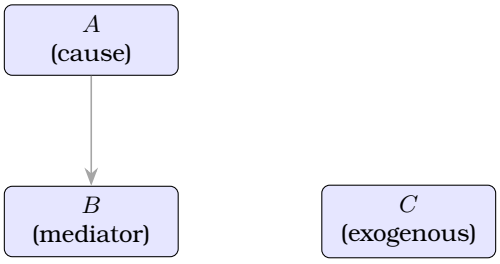
There are several strategies for translating the DAG into SHAP-values. Frye et al. (2021) propose **asymmetric Shapley values** or ASVs. Here, all paths through the lattice where B precedes A are given a weight of 0, meaning they do not affect the SHAP-values. In our example, this leaves 3 out of the 6 paths used in TreeSHAP. Table 3 shows the computations.

Figure 7: SHAP Values



Notes: Panel (a) gives $\phi_A = 2.5$, $\phi_B = 2.0$ and $\phi_C = 1.5$. Panel (b) shows the six per-permutation marginal contributions as grey points, with a box drawn at the hinges and a line at the median; ties are offset vertically so that all six are visible. Note the expanded scale in panel (b). With only six permutations the hinges fall on the smallest and largest values, so each box spans the full range and the whiskers have zero length; here the median also happens to equal the mean for all three features.

Figure 8: A Hypothetical DAG



4.3 Causal SHAP-Values

A different approach can be found in Heskes et al. (2020), who introduce **causal SHAP-values** (CSVs). Their approach relies on the *do*-operator introduced by Pearl (2000). That operator concerns the explicit *manipulation* of a particular feature. Accordingly, Heskes et al. (2020) redefine the prediction function as

$$\mathbb{E}[f(X)|do(X = \boldsymbol{x}^*)] \tag{5}$$

Under the DAG from Figure 8, the following holds:

1. Intervening on A alone ($do(A = 1)$): Because A causes B, setting $d(A = 1)$ causes the distribution of B to shift according to $p(B|do(A = 1)) = p(B|A = 1)$. Consequently, A is credited with both its *direct effect* and its *indirect effect* through the mediator B.

2. Intervening in B alone ($do(B = 1)$): By do-calculus rules, intervening on the downstream mediator B cuts the incoming arrow from A. The upstream root cause A remains at its marginal observational distribution $p(A)$. B receives c redit only for its direct effect—there is no back-propagation of credit to A.

CSV computations require some additional information in the form of a structural causal model (SCM) showing how A affects B. Let us assume the following SCM:

1. $p(A = 1) = 0.5$ and $p(C = 1) = 0.5$, i.e., independent coin flips determine the values of A and C.

2. We assume that A determines B with a high probability. Specifically,

$$\begin{aligned} A = 1 \quad &\Rightarrow \quad p(B = 1|do(A = 1)) = 0.8 \\ A = 0 \quad &\Rightarrow \quad p(B = 1|do(A = 0)) = 0.2 \end{aligned}$$

   The baseline marginal for B is $p(B = 1) = p(B = 1|do(A = 1)) \cdot p(A = 1) + p(B = 1|do(A = 0)) \cdot p)A = 0) = 0.8 \cdot 0.5 + 0.2 \cdot +0.5 = 0.5$.

We now recompute the lattice nodes for the constellation $\boldsymbol{x}^* = (A = 1, B = 1, C = 1)$. We start with the baseline: $v_{Causal}(\emptyset) = 4.0$. Next, we consider the intervention on B alone. Here, it is important to recognize there are no effects on A and C. Both features remain at their priors. When we average over the independent distributions of A and C, we get $v_{Causal}(\{B\}) = 5.6$.

Things are more complex when considering an intervention on A alone. Setting $do(A = 1)$ pushes downstream into B: $B = 1$ with probability 0.8 and $B = 0$ with probability 0.2. We calculate the expected output when $A = 1$, averaging over B and C. Specifically,

$$v_{Causal}(\{A\}) = 0.8 \cdot v(\{A, B\}) + 0.2 \cdot v(\{A, \bar{B}\})$$

Table 3: An Asymmetric Shapley-Value Example

		Marginal Steps		
Index	Permutation	A	B	C
1	$\emptyset \to \{A\} \to \{A,B\} \to \{A,B,C\}$	$\Delta_A(\emptyset) = 2.2$	$\Delta_B(\{A\}) = 2.0$	$\Delta_C(\{A,B\}) = 1.8$
2	$\emptyset \to \{A\} \to \{A,C\} \to \{A,B,C\}$	$\Delta_A(\emptyset) = 2.2$	$\Delta_B(\{A,C\}) = 2.4$	$\Delta_C(\{A\}) = 1.4$
3	$\emptyset \to \{B\} \to \{A,B\} \to \{A,B,C\}$	Invalid	Invalid	Invalid
4	$\emptyset \to \{B\} \to \{B,C\} \to \{A,B,C\}$	Invalid	Invalid	invalid
5	$\emptyset \to \{C\} \to \{A,C\} \to \{A,B,C\}$	$\Delta_A(\{C\}) = 2.4$	$\Delta_B(\{A,C\}) = 2.4$	$\Delta_C(\emptyset) = 1.2$
6	$\emptyset \to \{C\} \to \{B,C\} \to \{A,B,C\}$	Invalid	Invalid	Invalid
Average		$\phi_A = 2.27$	$\phi_B = 2.27$	$\phi_C = 1.47$

Notes: The paths indexed as 3, 4, and 6 receive 0 weights because B comes before A, which is inconsistent with the DAG.

From Table 1 we know that $v(\{A,B\}) = 8.2$. To get to $v(\{A,\bar{B}\})$ we compute

$$0.5 \cdot f(A=1,B=0,C=0) + 0.5 \cdot f(A=1,B=0,C=1)$$

Since $B = 0$ is counterfactual to $\boldsymbol{x}^*$, we have to return to the trees in Table 1. For the profile $A = 1, B = 0, C = 0$, Tree 1 yields $\hat{y} = 10.0$, with the remaining trees yielding predictions of $\hat{y} = 4.0$, $\hat{y} = -2.0$, $\hat{y} = 2.0$, and $\hat{y} = 2.0$, respectively. The average is 3.2. For the profile $A = 1, B = 0, C = 1$, the predictions are 10.0, 4.0, -2.0, 8.0, and 5.0, respectively. Those average to 5.2. Hence, $v(\{A,\bar{B}\} = 0.5 \cdot 3.2 + 0.5 \cdot 5.2 = 4.2$. Consequently, $v_{Causal}(\{A\}) = 0.8 \cdot 8.2 + 0.2 \cdot 4.2 = 7.4$.

At this point, it is important to note that CSV attributes more power to A than TreeSHAP: the prediction is 7.4 versus 6.2. The difference is that, in our setup, knowing $do(A = 1)$ means that B is more likely as well. TreeSHAP ignores that information.

We have to still consider one more possibility: intervention on A and B together. This is quite easy because B no longer varies. Hence,

$$v_{Causal}(\{A,B\}) = f(A=1,B=1) = 8.20$$

How do we now get to the direct and indirect effects? We first create the Δ when A is introduced via $do(A = 1)$,

$$\Delta_A^{Causal}(\emptyset) = 7.4 - 4.0 = 3.4$$

Table 4: Causal SHAP versus TreeSHAP

Feature	Standard SHAP Step	Causal SHAP Step	Comment
A (Cause)	2.20	3.40	2.20 (direct) + 1.20 (indirect via B)
B (Mediator)	2.00	0.80	Pure direct effect left over
Total ($\{A, B\}$	4.20	4.20	Combined attribution remains preserved

Per Heskes et al. (2020), the direct effect of A is

$$\phi_A^{Direct} = v(\{A, \text{prior}(B)\}) - v(\emptyset) = 6.2 - 4.0 = 2.2$$

This is $v(\{A\}$ from Table 1. The indirect effect is

$$\phi_{A \to B}^{Indirect} = v_{Causal}(\{A\}) - v(\{A, \text{prior}(B)\}) = 7.4 - 6.2 = 1.2$$

Notice that the direct and indirect effects add up to 3.4, which is the $\Delta_A^{Causal}(\emptyset)$.

After all the arithmetic is done, we observe that CSF tends to give a quite different picture then TreeSHAP. The differences are summarized in Table 4.

4.4 Evaluation

SHAP provides an axiomatic, permutation-invariant allocation of credit that avoids reliance on greediness or arbitrary oath-dependence of sequential learners Lundberg and Lee (2017). This makes it a natural candidate for tree ensembles where individual base learners already permute splitting orders across the forest Lundberg et al. (2020). Nevertheless, its game-theoretic foundation solves a functional accounting problem, not an epistemic or causal one. Standard SHAP values evaluate models on physically impossible or out-of-sample feature combinations, making them poor candidates for recourse Kumar et al. (2020), Merrick and Taly (2019), and Slack et al. (2020).[2] These are valid criticisms that necessitate care in interpretation. Standard SHAP values are a true-to-the-model audit of what the model has learnt. If we want to go beyond this, then causal assumptions must flow into the analysis and guide the construction of permutations Frye et al. (2021), Heskes et al. (2020), and Janzing et al. (2019).

[2] In interpretable machine learning, recourse references actions that would change the outcome. For instance, "your credit score *will* rise if you lower your debt-to-income ratio."

References

Biecek, P., & Burzykowski, T. (2021, February 17). *Explanatory Model Analysis: Explore, Explain and Examine Predictive Models* (1st ed.). Chapman and Hall/CRC.

Frye, C., Rowat, C., & Feige, I. (2021, December 20). *Asymmetric Shapley Values: Incorporating Causal Knowledge into Model-Agnostic Explainability*. arXiv: 1910.06358 [stat.ML].

Gigerenzer, G., & Goldstein, D. G. (1996). Reasoning the Fast and Frugal Way: Models of Bounded Rationality. *Psychological Review*, *103*(4), 650–669.

Gosiewska, A., & Biecek, P. (2019). *Do Not Trust Additive Explanations* (3).

Green, L., & Mehr, D. R. (1997). What Alters Physicians' Decisions to Admit to the Coronary Care Unit? *The Journal of Family Practice*, *45*(3), 219–226.

Heskes, T., Sijben, E., Bucur, I. G., & Claassen, T. (2020, November 3). *Causal Shapley Values: Exploiting Causal Knowledge to Explain Individual Predictions of Complex Models*. arXiv: 2011.01625 [cs.AI].

Janzing, D., Minorics, L., & Blöbaum, P. (2019, November 27). *Feature Relevance Quantification in Explainable AI: A Causal Problem*. arXiv: 1910.13413 [stat.ML].

Kumar, I. E., Venkatasubramanian, S., Scheidegger, C., & Friedler, S. (2020, June 30). *Problems with Shapley-Value-Based Explanations as Feature Importance Measures*. arXiv: 2002.11097 [cs.AI].

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From Local Explanations to Global Understanding with Explainable AI for Trees. *Nature Machine Intelligence*, *2*(1), 56–67.

Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Proceedings of the 31st International Conference on Neural Information Processing Systems*, 4768–4777.

Merrick, L., & Taly, A. (2019). *The Explanation Game: Explaining Machine Learning Models Using Shapley Values* (3).

Pearl, J. (2000). *Causality: Models, Reasoning, and Inference*. Cambridge University Press.

Shapley, L. S. (1953). A Value for N-Person Games. In H. W. Kuhn & A. W. Tucker (Eds.), *Contributions to the Theory of Games* (pp. 303–317). Princeton University Press.

Slack, D., Hilgard, S., Jia, E., Singh, S., & Lakkaraju, H. (2020, February 3). *Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods*. arXiv: 1911.02508 [cs.LG].

Staniak, M., & Biecek, P. (2019). Explanations of Model Predictions with live and breakDown Packages. *The R Journal*, *10*(2), 395.

Tversky, A. (1972). Elimination by Aspects: A Theory of Choice. *Psychological Review*, *79*(4), 281–299.